

LESSON 9-4 • SESSION 1 • PRACTICE • FORM A

Substitute and Solve

Learning target – I can substitute a value into an equation to find the matching output. • 6.EE.C.9

REMEMBER

- **Substitute** means put a known number in place of a letter. Then do the math and add units.
- With a fee, **multiply first**, then add: $c = 3n + 10$ with $n = 4$ is $3 \times 4 + 10 = 22$.
- Check that the answer is **reasonable** and answers the question asked.

WORKED EXAMPLE

A bakery makes 24 rolls each hour: $r = 24h$. Is 5 hours enough for 150 rolls?

Step 1 Substitute $h = 5$: $r = 24 \times 5 = 120$ rolls.

Step 2 Compare: $120 < 150$.

Answer No, 5 hours makes only 120 rolls

WORDS substitute (*sustituir*) • equation (*ecuación*) • reasonable (*razonable*)

PRACTICE

show your work • label units

1. Substitute to find the output.

a. $y = 6x$; $x = 12$ _____

b. $c = 5h + 10$; $h = 4$ _____

c. $y = 9x + 2$; $x = 5$ _____

d. $y = 7x + 3$; $x = 10$ _____

2. **Candle Workshop.** A workshop pours 18 candles every hour. A store orders 900 candles.

a. Write an equation for the candles y made in x hours. _____

b. Find the candles made in 12 hours. _____ candles

c. Is a 7-hour day enough for the order? Explain. _____

3. **Cookie Trays.** A bakery bakes 24 cookies on every tray. Write an equation for the cookies c on t trays. Find the cookies on 16 trays.

Hint Write the rule, then put in the number of trays.

Show your work.

Answer _____ cookies

4. **Steady Swimmer.** Jordan swims 50 meters every minute. Write an equation for the meters m in n minutes. Find the meters in 9 minutes.

Hint Replace the letter for minutes with the number.

Show your work.

Answer _____ meters

5. **Sandbag Crew.** A crew fills 8 sandbags every minute. After 45 minutes, the crew says it filled 360 sandbags. Does that claim fit the rate?

Hint Substitute the time, then compare with the claim.

Show your work.

Fits? _____

- 6. Skateboard Shop.** A shop builds 2.5 skateboards each hour. Write an equation for the boards b in h hours. Find the boards built in 24 hours.

Hint Multiply the decimal rate by the hours.

Show your work.

Answer _____ boards

- 7. With a Fee.** A rental costs $c = 14h + 25$ dollars for h hours. Find the cost for 6 hours.

Hint Multiply first, then add the fee.

Show your work.

Answer _____ dollars

- 8. Print Shop.** A shop charges $c = 3n + 20$ dollars for n posters.

- a. Find the cost of 15 posters. _____ dollars
- b. Find the cost of 30 posters. _____ dollars
- c. Explain why doubling the posters does not double the cost.

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Omar's Work.** A tap fills 6 buckets each minute. Omar wrote $b = 6m$. Then he said 30 minutes would fill 36 buckets. Explain his mistake and find the correct number.

TEST PRACTICE

choose the best answer

- 10.** For $y = 6x$, find y when $x = 8$.

A 14 **B** 48 **C** 68 **D** 86

- 11.** In an equation with a fee, like $c = 3n + 10$, you –

A add the fee first, then multiply **B** multiply first, then add the fee

C multiply the fee by n **D** skip the fee

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

84 Z	60 W	59 J	234 Q	40 X	110 R	216 A	450 S	109 E	384 N
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#2b	#3	#4	#6	#7	#8b
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____